MANDATORY DISCLOSURE AND HOUSING MARKET DYNAMICS

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SciencesPo.

Motivation

How Does Mandatory EPC Disclosure Reshape Housing Prices?

Three-step policy sequence in France

January 2006. EPC at the point of transaction — not visible during the search.

January 2011. EPC in ads — non-binding, no penalties, minimal enforcement.

October 2021. **Mandatory, legally binding, with penalties.**
Dual focus: GHG emissions *and* energy consumption.
Mandatory display in all online listings.

Why it matters:

Promoting energy-efficient choices.

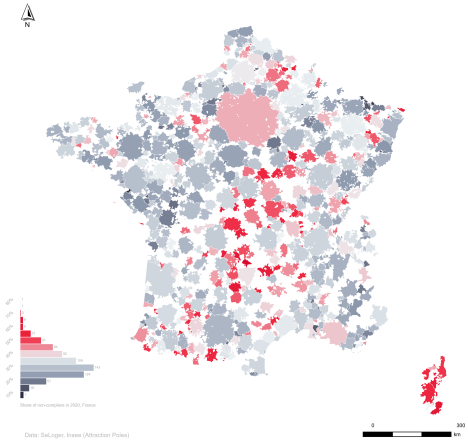
Encouraging renovations for inefficient homes.

Meeting national and international climate goals.

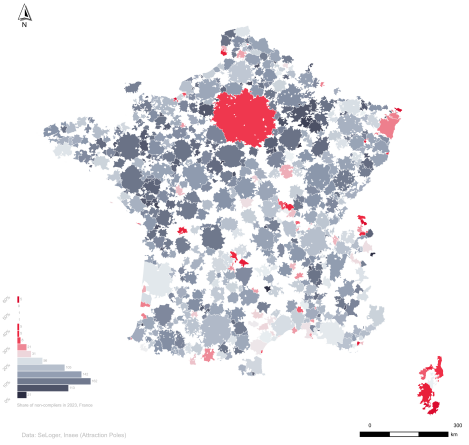
▸ Data sample ▸ Data description

Core Question. How does mandatory EPC disclosure affect real-estate pricing dynamics, seller disclosure decisions, and information transmission?

Unraverling of Energy Performance Score. Unraveling of energy performance should occur as long as (i) information is cheap and (ii) sellers are self-aware — *Grossman 1981 and Milgrom 1981*

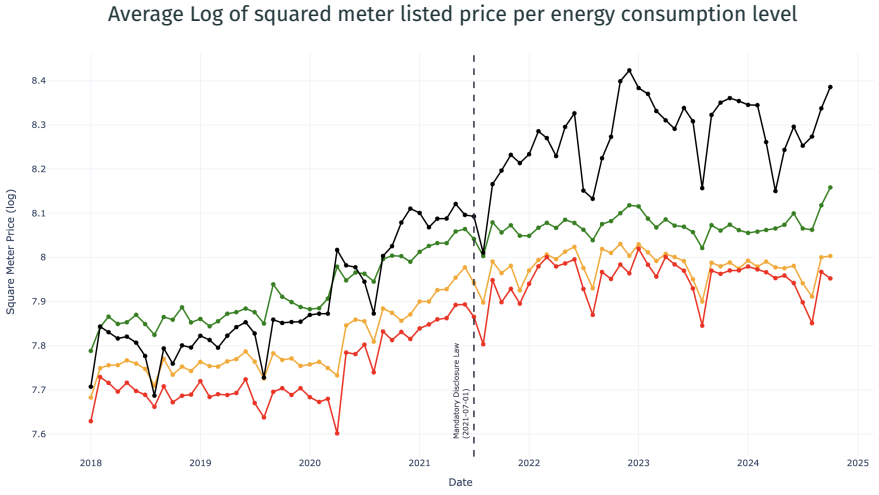


Share of listings posted without EPC - 2020



Share of listings posted without EPC - 2023

Motivation - Premiums over time



Contribution

Information Disclosure

Mandatory EPC disclosure **causally reshapes housing prices**.

Milgrom and Roberts 1986, Winston 2008, Dranove and Jin 2010

Asymmetric Price Effects

Brown discounts are large and persistent; green premia are limited — and both are heterogeneous across property types.

Myers 2020, Davis, Martinez, and Taboada 2020, Myers, Puller, and West 2022

Unraveling Failure

Non-disclosure remains profitable even under binding regulation — Grossman–Milgrom unraveling fails in the field.

Grossman 1981, Milgrom 1981

Methodology

ML-adjusted DiDisc at the France–Belgium border + **nationwide event-study DiD** under spatial heterogeneity.

De Chaisemartin et al. 2022, Picchetti, Pinto, and Shinoki 2024, Cattaneo, Idrobo, and Titiunik 2024, Noack, Olma, and Rothe 2025

Causal Effect of Mandatory Disclosure

Leveraging Heterogeneous EPC Adoption in Europe

🇫🇷 France

Early Phase

2011

✗ Non-binding, no penalties
✗ Basic energy bill method
✗ No visibility requirements

Effective Phase

2021 — Treatment

✓ Legally binding, steep penalties
✓ Dual focus: GHG + consumption
✓ Mandatory label in online ads

🇧🇪 Belgium (Flanders & Wallonia)

Early Phase

2009 / 2011

✗ Non-binding, low fines
✗ Basic EPC labels only
✗ Minimal enforcement

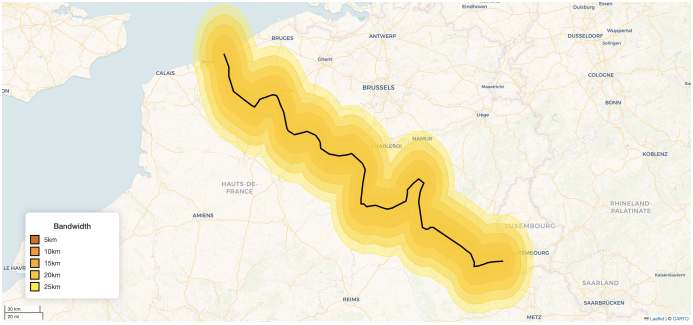
Effective Phase

2015 — Control Treatment

✓ Legally binding, higher fines
✓ Comprehensive EPC methodology
✓ Mandatory label in online ads

Identification at a glance

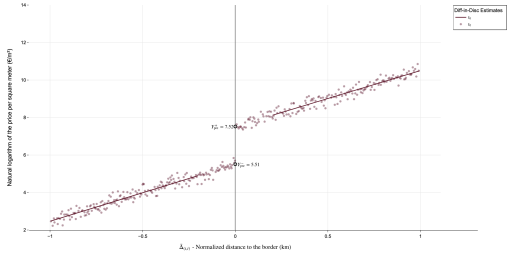
The border provides **quasi-random exposure to mandatory disclosure** while the **staggered adoption of mandatory disclosure allows us to net out permanent cross-country differences** unrelated to disclosure.



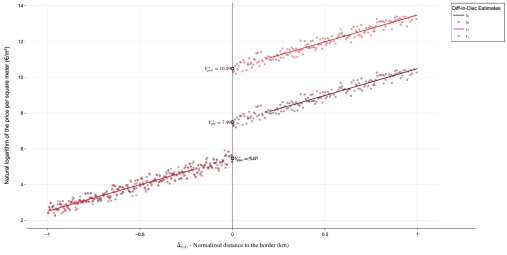
- ☞ **Exploit staggered disclosure** rules across borders.
- ☞ **Use geographic proximity** to ensure comparability.
- ☞ **Diff-in-Disc: compare price discontinuities at the border** before and after the French reform to isolate the causal effect of disclosure.

Diff-in-Disc: Removing Time-Invariant Confounders

Key Intuition. *The pre-period discontinuity can be estimated and subtracted from the post-period discontinuity, removing time-invariant confounders.*



Pre-reform (t_0): baseline gap from permanent country-level differences

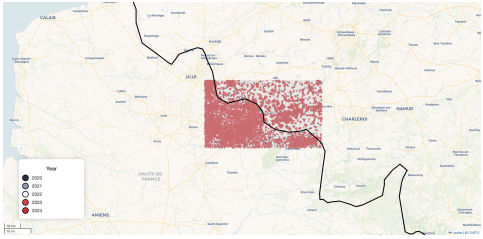


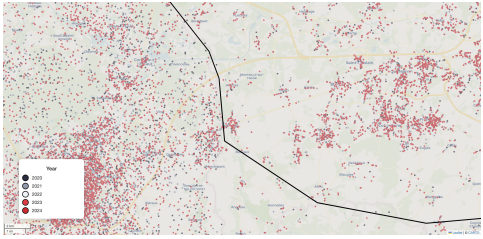
Post-reform (t_1): gap shifts — the difference $\hat{\tau}_{t_1} - \hat{\tau}_{t_0}$ is the causal treatment effect

$$\hat{\tau}^{\text{RDDisc}}(t) = \hat{\tau}_t - \hat{\tau}_{\text{baseline}}, \qquad t \neq \text{baseline}$$

Geo-locating listings around the border cutoff

Figure 1: Spatial distribution of geo-localized listed properties over time





Overview. *Collecting listings' location strategy*

- Run ElasticSearch Algorithm to identify precise location given reported address (Logitude/Latitudes).
- Restrict attention to subregions around the border and compute shortest distance in km to the border per listing.

▸ Fuzzy Location ▸ Data Description ▸ Descriptive Statistics

Estimation Challenge

Goal. Identify the causal effect of EPC disclosure by comparing listing prices across the France–Belgium border over time.

Why not compare raw prices directly?

- ✗ **Structural heterogeneity.** Housing characteristics differ across the border (even though stable over time)
- ✗ **Spatial heterogeneity.** Non-linear price gradients near the border confound the discontinuity.
- ✗ **Sample Size.** RDD requires biding to observations close to the border and at different energy consumption levels.

Solution. First remove predictable price variation via ML, then apply a geographic difference-in-discontinuities on adjusted prices.

▸ Spatial heterogeneity ▸ Balance table

Estimation Strategy — Algorithm

Solution. First remove predictable price variation via ML, then apply a geographic difference-in-discontinuities on adjusted prices.

Price Adjustment

Data-driven bandwidth — restrict to border listings (Calonico, Cattaneo, and Farrell 2020).

$$h^* = \max_t(h_t) \quad \forall t \in \{2019, \ldots, 2023\}$$

Cross-fitted ML residualisation — estimate conditional mean prices on each side of the border over each period (Noack, Olma, and Rothe 2025):

$$\eta_0(z_i) = \tfrac{1}{2}\left(\mu_0^+(z_i) + \mu_0^-(z_i)\right), \qquad \mu_0^*(z_i) = \mathbb{E}[Y_i \mid \Delta_i^* = 0,\ Z_i = z]$$

$$* \in \{+, -\}$$

Adjusted prices — net out spatial and structural heterogeneity over each time period:

$$\tilde{Y}_i(h^*) = Y_i(h^*) - \hat{\eta}_0(h^*, Z_i)$$

Causal Identification

Local linear RDD — estimate the border discontinuity on adjusted prices at each t:

$$\tau_t = \lim_{\delta \downarrow 0} \mathbb{E}\left[\tilde{Y}_{it} \mid \Delta_{it}^* = \delta\right] - \lim_{\delta \uparrow 0} \mathbb{E}\left[\tilde{Y}_{it} \mid \Delta_{it}^* = \delta\right]$$

$$\forall t \in \{2019, \ldots, 2023\}$$

Geographic difference-in-discontinuities — difference border gaps around the reform to recover dynamic treatment effects (Picchetti, Pinto, and Shinoki 2024):

$$\tau^{\text{RDDisc}}(t) = \tau_t - \tau_{\text{baseline}}, \qquad t \neq \text{baseline}$$

Full Sample Estimate

Independent Variable:	Natural log of price per square meter - €/m^2
Running Variable:	Normalized distance to the border (France/Belgium) - Δ^*

	Panel D: Full Sample				
Period	2019	2020	2021	2022	2023
First Stage Method: $\tilde{Y}_i(h^*,t) = Y_i(h^*,t) - \hat{m}_{0S}(h^*,Z_i,t)$	(RandomForest)	(RandomForest)	(RandomForest)	(LightGBM)	(LightGBM)
$\hat{\tau}_t$	0.344***	0.357***	0.334***	0.359***	0.347***
	(0.0126)	(0.0141)	(0.0125)	(0.0124)	(0.0113)
Robust Bias-Corrected CI	[0.32, 0.37]	[0.33, 0.38]	[0.31, 0.36]	[0.33, 0.38]	[0.32, 0.37]
Common Optimal Bandwidth (h^*)	10.430	10.430	10.430	10.430	10.430
$\hat{\tau}_t^{\text{DIDisc}} = \hat{\tau}_t - \hat{\tau}_{\text{baseline}}$	+1.1pp	+2.3pp	+0.0pp	+2.6pp	+1.3pp

Bandwidth Position	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Number of Observations	35 731	60 528	31 720	46 860	36 488	50 605	39 438	63 104	46 406	78 633
Number of Unique Obs.	35 731	60 528	31 720	46 860	36 488	50 605	39 438	63 104	46 406	78 633
Number of Effective Obs.	4 918	10 043	4 269	7 798	5 160	9 170	5 824	11 458	6 860	15 108

▸ Full Sample Fit

EPC Specific Estimates - Energy Intensive Sample

Independent Variable:	Natural log of price per square meter - €/m^2
Running Variable:	Normalized distance to the border (France/Belgium) - Δ^*

	Panel A: Energy Intensive				
Period	2019	2020	2021	2022	2023
First Stage Method: $\tilde{Y}_i(h^*,t) = Y_i(h^*,t) - \hat{m}_{0s}(h^*,Z_i,t)$	(RandomForest)	(RandomForest)	(RandomForest)	(RandomForest)	(RandomForest)
$\hat{\tau}_t$	0.319***	0.320***	0.361***	0.253***	0.242***
	(0.0421)	(0.0565)	(0.0382)	(0.0318)	(0.0281)
Robust Bias-Corrected CI	[0.24, 0.40]	[0.21, 0.43]	[0.29, 0.44]	[0.19, 0.32]	[0.19, 0.30]
Common Optimal Bandwidth (h^*)	11.256	11.256	11.256	11.256	11.256
$\hat{\tau}_t^{\text{DIDisc}} = \hat{\tau}_t - \hat{\tau}_{\text{baseline}}$	-4.2pp	-4.2pp	+0.0pp	-10.8pp*	-11.9pp*

Bandwidth Position	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Number of Observations	16 194	7 329	13 236	5 247	14 955	6 378	16 829	9 782	18 584	11 471
Number of Unique Obs.	16 194	7 329	13 236	5 247	14 955	6 378	16 829	9 782	18 584	11 471
Number of Effective Obs.	2 612	698	2 267	546	2 629	718	3 057	1 219	3 412	1 614

▸ Intensive Fit ▸ Balanced Fit ▸ Efficient Fit

EPC Specific Estimates - Energy Efficient Sample

Independent Variable:	Natural log of price per square meter - €/m^2
Running Variable:	Normalized distance to the border (France/Belgium) - Δ^*

	Panel B: Energy Balanced				
Period	2019	2020	2021	2022	2023
First Stage Method: $\tilde{Y}_i(h^*,t) = Y_i(h^*,t) - \hat{\eta}_{0S}(h^*,Z_i,t)$	(LightGBM)	(LightGBM)	(LightGBM)	(LightGBM)	(LightGBM)
$\hat{\tau}_t$	0.260***	0.313***	0.308***	0.308***	0.292***
	(0.0155)	(0.0177)	(0.0166)	(0.0157)	(0.0142)
Robust Bias-Corrected CI	[0.23, 0.29]	[0.28, 0.35]	[0.27, 0.34]	[0.28, 0.34]	[0.26, 0.32]
Common Optimal Bandwidth (h^*)	11.020	11.020	11.020	11.020	11.020
$\hat{\tau}_t^{\text{DIDisc}} = \hat{\tau}_t - \hat{\tau}_{\text{baseline}}$	-4.8pp*		+0.5pp	+0.0pp	+0.0pp	-1.6pp

Bandwidth Position	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Number of Observations	17 673	50 215	16 369	38 829	18 999	41 146	19 899	50 135	23 996	63 584
Number of Unique Obs.	17 673	50 215	16 369	38 829	18 999	41 146	19 899	50 135	23 996	63 584
Number of Effective Obs.	2 398	9 502	2 061	7 283	2 620	8 519	2 847	10 502	3 482	13 968

▸ Intensive Fit ▸ Balanced Fit ▸ Efficient Fit

EPC Specific Estimates - Energy Balanced Sample

Independent Variable:	Natural log of price per square meter - €/m^2
Running Variable:	Normalized distance to the border (France/Belgium) - Δ^*

	Panel C: Energy Efficient				
Period	2019	2020	2021	2022	2023
First Stage Method: $\tilde{Y}_i(h^*,t) = Y_i(h^*,t) - \hat{m}_{0s}(h^*,Z_i,t)$	(LightGBM)	(Lasso)	(Lasso)	(Lasso)	(RandomForest)
$\hat{\tau}_t$	0.181**	0.219***	0.218***	0.247***	0.207***
	(0.0595)	(0.0493)	(0.0509)	(0.0470)	(0.0393)
Robust Bias-Corrected CI	[0.06, 0.30]	[0.12, 0.32]	[0.12, 0.32]	[0.15, 0.34]	[0.13, 0.28]
Common Optimal Bandwidth (h^*)	13.981	13.981	13.981	13.981	13.981
$\hat{\tau}_t^{\text{DIDisc}} = \hat{\tau}_t - \hat{\tau}_{\text{baseline}}$	-3.7pp	+0.0pp	+0.0pp	+2.9pp	-1.2pp

Bandwidth Position	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Number of Observations	1 864	2 984	2 115	2 784	2 534	3 081	2 710	3 187	3 826	3 578
Number of Unique Obs.	1 864	2 984	2 115	2 784	2 534	3 081	2 710	3 187	3 826	3 578
Number of Effective Obs.	334	1 072	371	959	390	1 157	413	1 200	656	1 370

▸ Intensive Fit ▸ Balanced Fit ▸ Efficient Fit

From the Border to the Whole Country

What we have so far. The France–Belgium border design gives us a clean causal estimate of mandatory disclosure — but only for properties located *near the border*, and only for those who disclose.

What we still need.

- ☞ Does it matter whether a seller **discloses** or not?
- ☞ Are effects **homogeneous across France**, or do local market conditions shape the response?

Next step. We exploit the full nationwide panel of listings, using a two-step event-study DiD that compares EPC-group price dynamics within local markets.

The Heterogeneous effect of information

Broadening the Scope

Objective. Estimate the **Average Treatment Effect** of mandatory EPC disclosure, accounting for **local heterogeneity** in France's response.

Approach

Compare price changes across three treated groups **fixing the same reference property** across **local markets**:

- Energy-**inefficient** properties
- Energy-**efficient** properties
- **Non-disclosing** properties

Baseline: energy **balanced** properties (least affected by disclosure)

Assumptions

1. **Parallel trends** — treated and baseline groups share common pre-policy price trends
2. **Differential intensity** — treated groups face *stronger* treatment exposure than balanced dwellings, all else equal

▸ Data Description ▸ Local Markets ▸ Compositional Change

Estimation Strategy — Baseline

Objective. Identify how mandatory EPC disclosure **shifts green premia**, using balanced listings as a counterfactual for complying and non-complying treated listings.

Step 1. EPC Hedonic Price Indices

Build EPC-specific hedonic price indices at the cluster level (INSEE Attraction Poles) from repeated cross-sectional listing data:

$$I(\text{EPC}, g, t) : \begin{cases} \ln(p_i)|_{(\text{EPC},g)} & = \gamma_0 + \sum_{k=1}^{K} \beta_k \cdot X_{ik} + \gamma_t \cdot T_i + \epsilon_i \\ I(\text{EPC}, g, t \neq 0) & = 100 \times \exp(\gamma_t) \\ I(\text{EPC}, g, t = 0) & = 100 \end{cases} \tag{1}$$

- g local market (INSEE cluster, attraction poles)
- T are publication time dummies
- X_{ik} listed characteristics (floor, # bedrooms, city FE...)

Step 2. Event-Study Diff-in-Diff

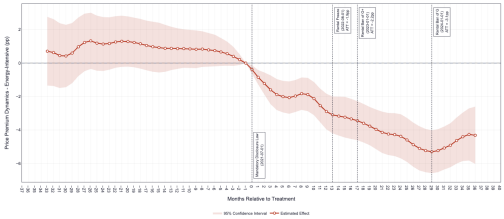
Construct a cluster-level panel of EPC-specific price evolutions and estimate an event-study DiD:

$$I_{gt} = \sum_{j=-m}^{n} \theta_j \cdot D_{g,t-j} + \alpha_g + \delta_t + \beta \cdot X_{gt} + \varepsilon_{gt} \tag{2}$$

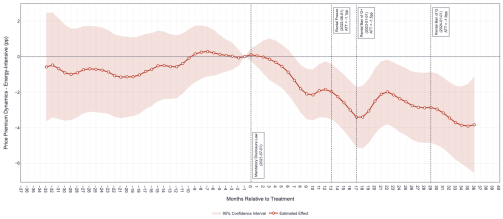
- α_g, δ_t cluster and time FE
- $D_{g,t-j}$ are event-time dummies
- $X_{g,t}$ is the volume of listings in cluster g at time t

Result 1: Brown Discount — Energy-Intensive vs. Balanced

TWFE: Balanced vs. Intensive — Panel HOUSE



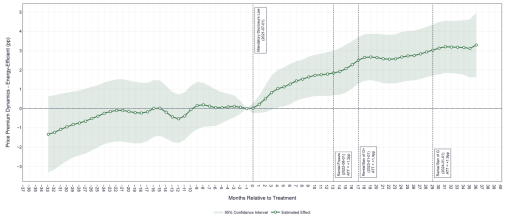
TWFE: Balanced vs. Intensive — Panel APARTMENT



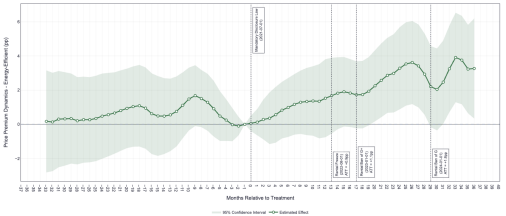
Brown Discount: −2 pp for apartments, −3 pp for houses relative to balanced properties — **both significant and stable post-reform.**

Result 2: Green Premium — Energy-Efficient vs. Balanced

TWFE: Balanced vs. Efficient — Panel HOUSE



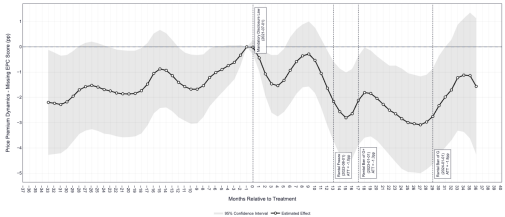
TWFE: Balanced vs. Efficient — Panel APARTMENT



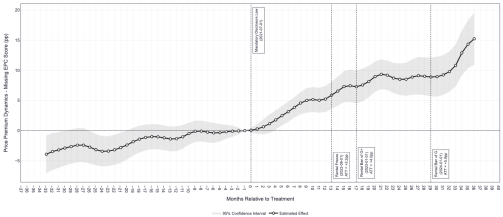
Green Premium: +3 pp for houses (statistically significant). **No significant effect for apartments** — location and amenities dominate over EPC information in the apartment segment.

Result 3: Non-Disclosure Premium — Unraveling Fails

TWFE: Balanced vs. Non-Compliers — Panel HOUSE



TWFE: Balanced vs. Non-Compliers — Panel APARTMENT



Non-Disclosure Premium (+6 pp for apartments): Under mandatory disclosure, non-disclosing apartments command a premium over balanced ones. **Sellers with uncertain or weak ratings optimally withhold under imperfect enforcement.**

▸ Anticipatory effects ▸ Panel statistics

Concluding Remarks

What we find. Mandatory EPC disclosure reshapes housing prices — but asymmetrically and incompletely.

- ✓ **Brown discount** is large and immediate: energy-inefficient dwellings lose up to 10 pp at the border, 2–3 pp nationwide.
- ✓ **Green premium** is limited and heterogeneous: significant for houses (+3 pp), absent for apartments.
- ✓ **Non-disclosure remains profitable**: non-disclosing apartments command a +6 pp premium — standard unraveling fails.

Why does non-disclosure persist?

- ☞ **Strategic non-disclosure:** Sellers with weak or uncertain EPC ratings optimally withhold when enforcement is imperfect.
- ☞ **Heterogeneous preferences:** Energy efficiency is valued differently across buyer segments, diluting the signal.
- ☞ **Rational ignorance:** Sellers uncertain about their own score have little incentive to disclose. Myers, Puller, and West 2022
- ☞ **Market segmentation:** In apartments, location and amenities dominate, crowding out EPC information.

Policy implication. Mandatory disclosure alone is insufficient — stricter enforcement and buyer education are needed to complete the informational transmission.

Thank You

PATRAS Bastien

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ANNEXES

Estimating Green Premiums Dynamics

Hedonic Pricing. *Decomposing property prices into individual characteristics.*

- Assumes the **price reflects the sum of values of each attribute** (e.g., location, area, amenities).
- Widely used in **real estate estimation** to provide accurate and detailed property valuations.

Why is Hedonic Pricing Effective for Green Premium estimation ?

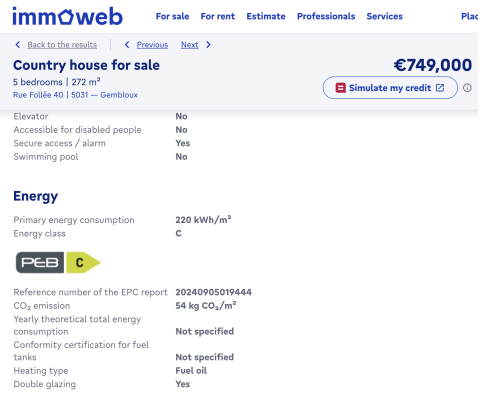
- Helps isolate the value and dynamics of **specific property features - EPC premiums -** from other information mechanisms (construction date, location, floor, heating system...).

Estimation. *Hedonic price estimates over time for cluster **g** and EPC* $\in$ { ***Efficient***, ***Balanced***, ***Inefficient***, ***Non-Compliers***}.

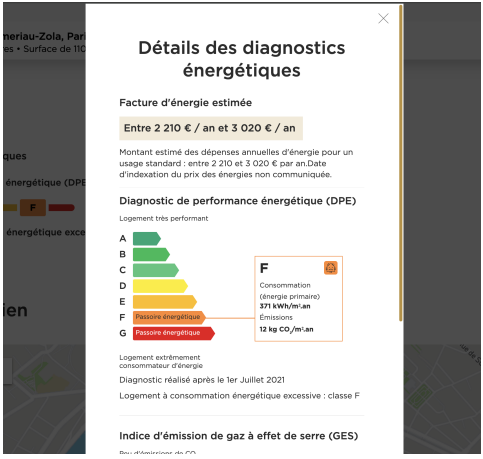
$$\begin{cases} \ln\left(\frac{P_i}{S_i}\right)\Big|_{(\text{EPC},g)} & = \gamma_0 + \sum_{k=1}^{K} \beta_k X_{ik} + \gamma_t \cdot T_i + \epsilon_i \\ \text{Index}_{(\text{EPC},g,t)} & = 100 \times \exp(\gamma_t) \\ \text{Index}_{(\text{EPC},g,t=0)} & = 100 \end{cases} \tag{3}$$

▸ Return

Website overview



Immoweb - PEB



Seloger Group - DPE

Return

Traditional RDD at the border cutoff

Traditional RDD Assumptions. *Between 2015 and 2021 France is not under stringent mandatory disclosure while it is the case for Belgium which create a discontinuity in price at multiple EPC levels.*

Assumption	Application to Border Context
Continuity	**Assumes housing prices are comparable near the border**, so any change in price at the border is attributed to the country-specific policy.
Local Randomization	Treats listed properties near the border as if they were **randomly placed on either side**, minimizing pre-existing differences in characteristics.
No Manipulation Around the Cutoff	Assumes **individuals do not strategically position themselves** based on policy advantages across the border.
Homogeneity of Treatment Effect	**Assumes similar treatment effect near a border side for all units.** Differences in amenities or historical factors on either side may affect price heterogeneity.
Smoothness of Covariates	Assumes **socio-economic variables do not change sharply at the border.** Cross-border differences in demographics and local economies could violate this assumption.
Consistency of Running Variable	**Requires accurate location measurement to determine whether a property is in France or Belgium.** Misclassifications weaken the treatment effect estimate.

Data sources

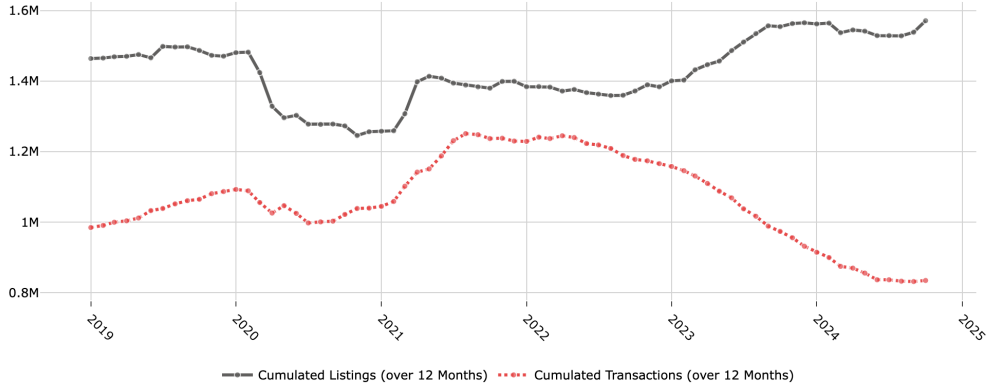
- **France:** SeLoger listings, 2018–2024 (≈10.3M listings)
- **Belgium:** Immoweb listings, border sample (≈275k listings)
- Listings posted by **professional real-estate agents**
- Rich information: price, surface, rooms, location, amenities, EPC

Market Coverage

- Platform coverage exceeds transactions
- Example (2024): 1.5M listings vs. 837k transactions (French notaries)
- De-duplicated across time and platforms ⇒ one dwelling once

▸ Return - Border ▸ Return - Nationwide ▸ Return - Introduction

Figure 2: Cumulated Listings and Transactions in the French Residential Market (2019-2024)



Sources: De-duplicated listings, SeLoger France (authors' own computations), Real Estate Reference Database, French Notaries – ADSN, DGFiP (MEDOC) - (Insee computations).

Limitations in Applying RDD to Border Contexts

Limitations. Self-selection and persistent policies may violate local randomization and confound treatment effects in analysis.

- ☞ **Selective Sorting and Self-Selection.** *Residents may self-select regions for economic advantages, violating the assumption of local randomization.*
- ☞ **Alternative Time-Invariant Policies.** *Persistent policies unrelated to recent border changes (e.g., tax regimes, regulations) may confound the treatment effect.*

Solution. Difference-in-Discontinuity ! Allows for discontinuities from compound treatment and time-invariant sorting in t_0, correctly identified as long as:

- ☞ Absence of **lagged sorting** or **treatment related sorting**
- ☞ **Absence of significant policy triggered in between periods** generating another discontinuity.

▸ Traditional RDD

Random Sampling of Fuzzy Location - Point(long, lat)

Issue. *Longitudes and latitudes are not always properly reported on SeLoger. Instead of dropping observations we follow an alternative solution*

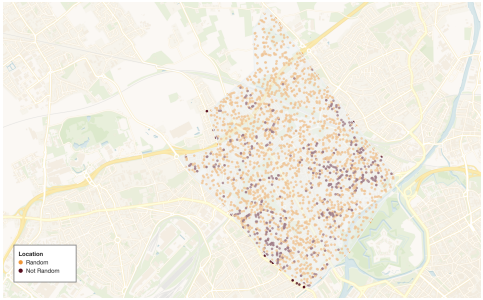
Solution. *Random Sampling of Fuzzy Location*

- ☞ **Fuzzy location:** The location is not precisely known but is constrained within a certain boundary (city geometry).
- ☞ **Random sampling:** Random point is selected from within the fuzzy boundary using probabilistic techniques.

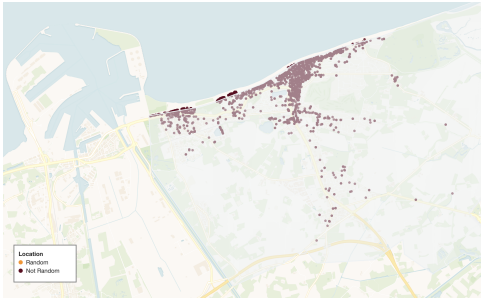
Objective. *Generating a plausible location from the given area when the exact coordinates are missing but the related city is known.*

▸ Return

Figure 3: Random Sampling of Fuzzy Location



France - $\bar{\Delta}^{c}_{\{i,t\}} = 20\text{km}$



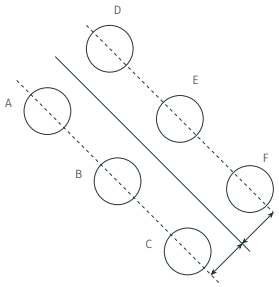
Belgium - $\bar{\Delta}^{c}_{\{i,t\}} = 20\text{km}$

Return

Spatial Heterogeneity

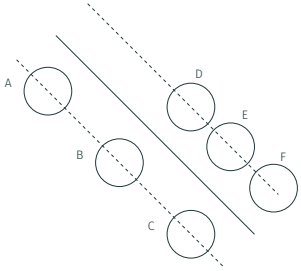
Uniform poles

Baseline case



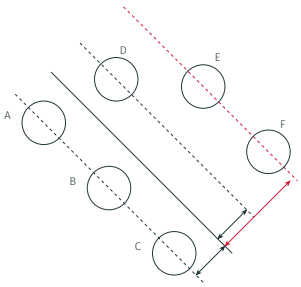
Latitudinal skew

Poles cluster on one side



Longitudinal skew

Asymmetric border gradient



◯ Attraction pole —— Administrative border - - - Bandwidth - - - Distorted gradient

▸ Return

Table 1: Coefficient of determination for various η_0 specifications at multiple time period (t) - Full Sample

Period (t)	Lasso Regression	Gradient Boosting (LightGBM)	Linear Regression	Random Forest
2019	24.6 %	22.6 %	-1.4e+27 %	29.4 %
2020	25.2 %	21.5 %	-2.4e+27 %	30.5 %
2021	26.8 %	23.7 %	-8.0e+26 %	31.3 %
2022	20.7 %	25.8 %	-8.9e+26 %	20.8 %
2023	23.8 %	26.6 %	-5.0e+26 %	25.4 %

Note: Observations are restricted to a bandwidth of 10.43km buffer around the border between France and Belgium. This distance buffer is estimated using data-driven optimal bandwidth as described in Calonico, Cattaneo, and Farrell 2020.

Return

Table 2: Coefficient of determination for various η_0 specifications at multiple time period (t) - Intensive sample

Period (t)	Lasso Regression	Gradient Boosting (LightGBM)	Linear Regression	Random Forest
2019	16.3 %	10.5 %	-2.2e+28 %	19.4 %
2020	15.9 %	8.2 %	-2.7e+28 %	18.7 %
2021	18.4 %	17.0 %	-2.4e+28 %	22.9 %
2022	16.9 %	12.6 %	-2.4e+28 %	20.1 %
2023	21.8 %	14.8 %	-1.2e+28 %	24.0 %

Note: Observations are restricted to a bandwidth of 11.256km buffer around the border between France and Belgium. This distance buffer is estimated using data-driven optimal bandwidth as described in Calonico, Cattaneo, and Farrell 2020.

Return

Table 3: Coefficient of determination for various η_0 specifications at multiple time period (t) - Balanced Sample

Period (t)	Lasso Regression	Gradient Boosting (LightGBM)	Linear Regression	Random Forest
2019	20.2 %	26.3 %	-3.1e+27 %	19.1 %
2020	20.2 %	25.3 %	-2.6e+27 %	22.1 %
2021	19.2 %	23.6 %	-4.4e+27 %	18.9 %
2022	18.4 %	25.0 %	-1.6e+27 %	19.7 %
2023	17.5 %	24.9 %	-6.7e+26 %	22.2 %

Note: Observations are restricted to a bandwidth of 11.020km buffer around the border between France and Belgium. This distance buffer is estimated using data-driven optimal bandwidth as described in Calonico, Cattaneo, and Farrell 2020.

Return

Table 4: Coefficient of determination for various η_0 specifications at multiple time period (t) - Efficient Sample

Period (t)	Lasso Regression	Gradient Boosting (LightGBM)	Linear Regression	Random Forest
2019	26.0 %	30.3 %	-8.5e+27 %	28.5 %
2020	28.3 %	26.7 %	-3.5e+28 %	23.4 %
2021	23.4 %	23.2 %	-4.2e+27 %	15.6 %
2022	24.1 %	22.3 %	-1.9e+28 %	23.3 %
2023	26.2 %	24.2 %	-1.8e+28 %	27.9 %

Note: Observations are restricted to a bandwidth of 13.981km buffer around the border between France and Belgium. This distance buffer is estimated using data-driven optimal bandwidth as described in Calonico, Cattaneo, and Farrell 2020.

Return

Les pistes pour desserrer les contraintes sur les passoires thermiques

IMMOBILIER

« Le diagnostic de performance énergétique sera simplifié, et son calendrier, adapté », s'est engagé le Premier ministre, Michel Barnier.

Les professionnels du secteur avancent leurs propositions.

Elsa Delaroy

Le compte à rebours est lancé. Dès le 1er janvier 2025, les logements classés G au titre du diagnostic de performance énergétique (DPE), c'est-à-dire les plus énergivores, ne pourront plus faire l'objet d'un bail de renouvellement. Dès le 1er janvier 2028, ce sera au tour des logements F. Puis, dès le 1er janvier 2034, ce sera le tour des E. Autant d'échéances qui ont été fixées par la loi Climat et résilience, promulguée en août 2021, pour lutter contre les passoires thermiques, ces logements qui consomment beaucoup d'énergie et émettent d'importantes quantités de gaz à effet de serre, et qui doivent progressivement disparaître du parc locatif.

Mais à mesure que ces échéances approchent, le nombre de logements concernés par ces interdictions ne cesse de grimper. Selon le dernier bilan publié par le ministère de la Transition écologique, ils étaient 4,8 millions en 2022, contre 4,3 millions en 2021. Et ce chiffre pourrait encore augmenter, car il ne prend pas en compte les logements qui ont été rénovés depuis 2022 mais qui ne sont pas encore passés à un DPE plus favorable.

Face à cette situation, le gouvernement a annoncé, dans sa déclaration de politique générale, le 1er octobre, qu'il allait « *simplifier le diagnostic de performance énergétique (DPE) et son calendrier, adapté* ». Mais sans donner plus de détails. Il a également indiqué qu'il allait « *mettre en place un choc de simplification pour le logement* ». Sans plus de précisions, là encore.

Pour y voir plus clair, les professionnels du secteur ont donc avancé leurs propositions. Dans un rapport publié ce mardi 8 octobre, la Fédération des promoteurs immobiliers (FPI), présidée par Alain Dinin, propose ainsi de « *mettre en place un moratoire de trois ans sur les interdictions de mise en location des logements classés F et G dans le parc locatif privé* ». Pour Alain Dinin, « *le Premier ministre a raison d'être très attentif à la question du logement, qui est la première préoccupation des Français. Nous avons des propositions à faire pour que les mesures annoncées soient efficaces et ne pénalisent pas davantage le pouvoir d'achat des ménages* ».



Au 1er janvier 2023, l'Observatoire de la rénovation énergétique dénombrait près de 1,5 million de passoires énergétiques (logements F et G) dans le parc locatif privé. *Photo Laurence Geai/opale.photo/AFA*

Rénovation urbaine, Valérie Létard, « *Le Premier ministre a raison d'être très attentif à la question du logement, qui est la première préoccupation des Français. Nous avons des propositions à faire pour que les mesures annoncées soient efficaces et ne pénalisent pas davantage le pouvoir d'achat des ménages* »

Parmi les mesures phares avancées par la FPI, celle d'un moratoire de trois ans sur les interdictions de mise en location des logements classés F et G dans le parc locatif privé. Pour Alain Dinin, « *le Premier ministre a raison d'être très attentif à la question du logement, qui est la première préoccupation des Français. Nous avons des propositions à faire pour que les mesures annoncées soient efficaces et ne pénalisent pas davantage le pouvoir d'achat des ménages* ».

Demande de clarification

Faire le tri et la réhabilitation effective des trésors dans l'immobilier, il peut rénover ce passer plusieurs années. Le temps de trouver une entreprise, le temps de réaliser les travaux, le temps de chauffer le logement durablement, d'obtenir les aides éventuelles, … « *Le coup de pinceau ne suffira pas* », indique le Premier ministre, Michel Barnier, à l'occasion de son discours de politique générale. « *Il faudra aussi de remettre en question – et même, être global des diagnostics et, au-delà, en cause de l'adapter […], avec les questions des copropriétés qui ont besoin de travaux et de financements* », a précisé un peu plus tard le ministre du Logement et de la Rénovation urbaine Valérie Létard.

« Deux tiers des logements aujourd'hui en étiquette G et qui ne sont toujours pas requalifiés sont en copropriété. »

VALÉRIE LÉTARD
Ministre du Logement et de la Rénovation urbaine

Faut-il alors réinviter l'accord des copropriétaires sur le sujet – alors qu'aux propriétaires de maison est mis à droits, à annuel la mission.

A la surprise générale, Michel Barnier a dit vouloir modifier le calendrier du diagnostic de performance énergétique

Volte-face sur l'interdiction de louer des passoires thermiques

Logement

Dans trois mois, environ 480 000 logements devraient être retirés du marché de la location. Si le calendrier du diagnostic de performance énergétique (DPE) était effectivement assoupli, Michel Barnier suivrait le PS, qui a déposé mi-septembre une proposition de loi pour repousser le délicat de trois ans.

Sandra Marhoud

« LE DIAGNOSTIC DE PERFORMANCE énergétique sera simplifié et son calendrier sera adapté. » Une petite phrase et tout un monde est en émoi. À peine une heure après le début de sa déclaration de politique générale, mardi, le Premier ministre aborde l'épineuse question du logement, un sujet qu'il connaît bien pour avoir été ministre du dernier set « la première poste de dépenses des Français », il prend la parti d'annoncer quelques mesures concrètes, dont la réforme du controversé DPE, ce qui sous-tend en probable report de l'interdiction de mise en location des logements les plus énergivores. Une nouvelle, sans surprise, pour les députés socialistes, mais qui a plu à toute l'autre côté de l'hémicycle.

« Insécurité ». Côté face, les propriétaires bailleurs, les premiers concernés par le DPE, et les acteurs du logement se réjouissent de cet assouplissement éventuel du calendrier. Côté pile, les défenseurs de l'environnement et de la justice sociale dénoncent un énième un crève d'achat et un outil des politiques extrêmement tales. Pour rappel, après analyse d'un diagnostic immobilier, le logement ou l'immeuble est noté de A (moins autonome) à G (passoire thermique). Pour obliger les propriétaires à faire des travaux de rénovation, la loi Climat et résilience (août 2021) a introduit de mettre en location les logements dont l'étiquette énergétique est comprise entre F et G selon un échéancier progressif.

Au 1er janvier 2025, les passoires classées G sont censées ne plus pouvoir être indécentes. Ce sera ensuite le tour des logements F (au 1er janvier 2028), puis E (en 2034)... Trois catégories qui représentent une part non négligeable de l'habitat en France : Sur les 30 millions de résidences principales recensées en 2021 par l'Insee, près de 6,4 millions (7 sont des passoires thermiques (F et G) – soit 15,7% du parc –, dont 1,5 million sont locataires des habitats privés. Le risque de voir ces biens sortir du parc locatif, en pleine crise du logement, semble très important. Juste après la publication de la loi de 2020, le nombre de logements proposés à la location avait déjà chuté de 10 % lorsque de quelques mois et représentaient 17% des transactions, contrastaient les ventes.

« Les propriétaires privés ont conscience depuis longtemps de l'urgence climatique et de la nécessité de réagir. Néanmoins, une nouvelle fois, ils déplorent l'absence de moyens et de visibilité pour y parvenir », réagit Jean Perrin, dernier en « le premier poste de dépenses des Français », il prend la parti d'annoncer quelques mesures concrètes, dont la réforme du controversé DPE, ce qui sous-tend un

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